THE CRUTCH TEST: AUDITING THE COGNITIVE COST OF DIRECT ANSWERS IN SIMULATED STUDENT ENVIRONMENTS

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ABSTRACT

Tutoring systems built on large language models (LLMs) are increasingly trained with reinforcement learning against simulated students. In a randomized trial with about a thousand high-school students, unrestricted GPT-4 raised practice scores but lowered scores on a later exam taken without AI by 0.054 on a 0–1 scale, or 17% (Bastani et al., 2025). A tutor that gave only hints did not lower them. We call this loss the *crutch cost*. If simulated students do not pay it, rewards built on them favor tutors that simply hand over answers. We introduce the **Crutch Test**. It runs a student simulator through the trial's three conditions (no AI, hints, and answers), tests the simulated student without help on a new variant of each problem, and checks the results against pass gates set from the trial. In a 50-problem pilot with a Llama-3.1-8B-Instruct student and Claude Haiku 5.5 tutors, all three simulators that carry the dialog as text get the direction wrong: being told the answer raises their later test score by 0.06 to 0.15. A simulator that instead learns in its weights, only from tokens it wrote, removes this gain at both learning rates we tried. It does not reproduce the human loss, and no setting passes every gate at this scale. Code is available at `https://github.com/NickTrinh/crutch-test`.

1 INTRODUCTION

Tutors built on LLMs are now trained with reinforcement learning against simulated students. The tutor talks with a student model, and the student's score on a later test becomes the tutor's reward (Dinucu-Jianu et al., 2025; Scarlatos et al., 2025; Chang et al., 2026; Xu et al., 2026; Macina et al., 2026). Such a reward is only as good as the simulated student behind it. If the simulated student learns as much from being told an answer as from working it out, a tutor trained against it learns to tell.

Human students do not learn this way. In a field trial with high-school mathematics students, Bastani et al. (2025) gave one group access to unrestricted GPT-4 ("GPT Base"). This group's practice scores rose by 0.137, a 48% gain. Once access was removed, however, its exam scores fell by 0.054 on a 0–1 scale (SE 0.022), or 16.8% of the control group's mean of 0.321. A second group used a tutor prompted to give hints rather than answers ("GPT Tutor"). Its practice scores rose by 0.361, and its exam scores barely changed (-0.004, SE 0.013). Online experiments show the same drop in unaided performance after AI help (Liu et al., 2026; Shen & Tamkin, 2026), and current assistants tend to give more help than students need (Teo et al., 2026).

Current simulated students do not show this loss. In EduardoRL (Macina et al., 2026), the student's post-test hides the tutor's turns, so the score depends only on what the student wrote. Even so, tutors that hand over the solution in 83–93% of dialogs still raise the student's score on a new variant of the problem by $+0.13$ to $+0.14$. The authors call such handover reward-neutral and add binary reward gates to suppress it. Sherpa uses a separate checker to block turns that reveal the answer (Xu et al., 2026). Both measures fix the reward. Neither tests whether the simulated student learns like a person.

We build that test. The Crutch Test runs a simulated student through the trial's three conditions. It then measures the student's score on a new variant of each tutored problem, taken without help, and compares the conditions problem by problem against gates set from the trial. We test three *frozen*

simulators, whose weights never change, so everything they learn must be carried as text. We also test one candidate fix: a simulated student that learns in its weights (Sim_{LoRA}). It trains only on the tokens it wrote itself, so a student who copied the answer has less to learn from.

Our main contributions are:

- **A test grounded in a human trial.** Four gates, fixed in advance ($\mathcal{G}_1$–$\mathcal{G}_4$), turn the trial's effects into checks that a simulator passes or fails.
- **A pilot audit.** On 50 problems (150 dialogs and 9,600 student calls, none failed), all three frozen simulators fail the crutch-cost gate $\mathcal{G}_2$ in the wrong direction: answers raise their scores by $+0.06$ to $+0.15$.
- **A learning student.** Sim_{LoRA} removes this gain at both learning rates we tried, but it does not reproduce the human cost, and the sign of its effect depends on the learning rate. We analyze why.
- **An open harness.** It pins model hosts, logs every answer and its cost, and includes a \$0 mock server and a GPU runner that survives client disconnects. A 300-problem held-out run, with two extra arms (S, try first; W, wrong solution), two dialogs per arm and Holm's correction, is built and tested on the mock server.

2 Related work

Tutor RL against simulated students. TutorRL rewards a tutor when a frozen student solves the same problem with the dialog still in its context (Dinucu-Jianu et al., 2025). DPO-trained tutors (Scarlatos et al., 2025) and PEARL (Chang et al., 2026) also optimize tutors against simulated students or learned rewards. EduardoRL hides the tutor's turns and tests the student on a near-transfer variant, yet still finds that handing over the answer costs the tutor no reward (Macina et al., 2026). Sherpa blocks turns that reveal the answer (Xu et al., 2026). Optimizing a per-decision teaching metric can raise that metric while educators' ratings fall (Domínguez Figaredo & Fernández De la Cruz, 2026). All of these treat telling as a problem of reward design. We treat it as a problem of measuring the student.

Validating student simulators. Prior work checks whether simulated students produce realistic dialog turns (Scarlatos et al., 2026), fit and update individual students' responses under guidance (Yang et al., 2026), predict a learner's next action (Jiang et al., 2026), and show human-like learning curves in tutoring systems (Weitekamp et al., 2025). Experiments with LLM-simulated subjects behave like observational studies unless they are designed carefully (Lin et al., 2026). A recent synthesis calls for validating each stage of the tutoring loop, including the simulated student (Adak et al., 2026). As of October 16, 2026, we found no work that compares a simulator's treatment effects with those of a randomized trial.

Students whose weights change. Song et al. (2026) unlearn a model down to a novice level and let it relearn during tutoring. Their student trains on the coach's explanation and the gold answer with AdamW, and it is evaluated against its own base model. Sim_{LoRA} borrows the mechanics of test-time training (Sun et al., 2019) and of self-training on a model's own generations (Zelikman et al., 2022; Zuo et al., 2025; Zweiger et al., 2025). It differs in three ways: it trains on all of its own tokens without filtering, its update grows with how much it wrote, and it is tuned to reproduce a human learning cost rather than to raise accuracy.

Human evidence. People remember material they generate better than material they read (Slamecka & Graf, 1978). Unrestricted AI help lowered later unaided performance in a field trial (Bastani et al., 2025) and in online experiments (Liu et al., 2026; Shen & Tamkin, 2026). Only Bastani et al. (2025) also randomized a hint condition, so the target for hints rests on a single trial.

3 Methodology

3.1 Protocol

Each problem p has a near-transfer variant p' that the student never sees during tutoring. The student talks with a tutor in one of three arms $a \in \{N, H, T\}$ (Figure 1). In arm T (answers), the tutor uses

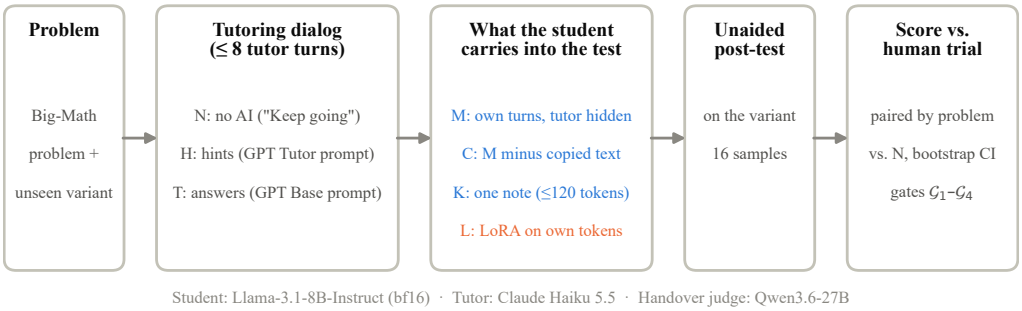


Figure 1: Overview of the Crutch Test. Each problem is tutored in three arms. A simulator is a rule for what the student carries from the dialog into a test on a new variant of the problem, taken without help. Each arm is compared with the no-AI arm problem by problem, and the differences are checked against gates set from the human trial.

the trial's GPT Base prompt. In arm H (hints), it uses the trial's GPT Tutor prompt. In arm N (no AI), the student receives the scripted turn "Keep going. Show your next step." as many times as the tutor spoke in arm H on that problem. Every dialog opens with the student asking "What is the answer?", the most common first message in the trial's GPT Base arm. Both tutor prompts end with a rule: discuss only this problem, and close the dialog when the student has no further questions. A dialog has at most eight tutor turns.

After the dialog, the student takes a post-test on p' without help. A *simulator view* v fixes what the student carries into this test. The score $s_{a,v}(p)$ is the fraction of K sampled answers that are correct, so every score lies between 0 and 1.

3.2 Simulator views

Sim_M keeps the student's own turns and hides the tutor's, as in EduardoRL (Macina et al., 2026). Sim_C goes further: it deletes every student sentence that shares at least half of its word 6-grams with a tutor turn, to remove copied text. This rule overlaps with what the handover judge looks for, so Sim_C serves only as a reference. Sim_K carries a single note of at most 120 tokens, which the student writes with the whole dialog in view. These three are the frozen simulators.

Sim_{LoRA} carries no transcript. Instead, it learns from the dialog in its weights. For a dialog d with tokens $x_{1:T}$, let $m_t = 1$ when token t belongs to a student turn other than the scripted opener. A fresh, seeded LoRA adapter θ (Hu et al., 2021) on the frozen base model takes four plain SGD steps on the summed loss over the student's own tokens:

$$\mathcal{L}_d(\theta) = \frac{1}{1000} \sum_{t=1}^{T} m_t \left(-\log p_\theta(x_t \mid x_{<t})\right), \qquad \theta_{k+1} = \theta_k - \eta \nabla_\theta \mathcal{L}_d(\theta_k), \quad k = 0, \ldots, 3. \tag{1}$$

The loss is summed rather than averaged, so the update grows with the amount of work the student did itself. Adam would rescale each coordinate and remove this effect. The learning rate is fixed before any test, using the N and H dialogs only. It is the rate whose median loss drop over dialogs is closest to 25%:

$$\eta^\star = \arg \min_{\eta \in \{0.03, 0.1, 0.3, 1, 3\}} \left| \operatorname*{median}_d \left(1 - \mathcal{L}_d(\theta_4)/\mathcal{L}_d(\theta_0)\right) - 0.25 \right|. \tag{2}$$

The adapted student is then tested without a transcript on three problems: the variant ($K = 16$), the same problem restated ($K = 8$), and an unrelated problem ($K = 8$). The unrelated problem is another dialog's variant, the same for every arm. The frozen base model takes the same tests.

3.3 CONTRASTS AND GATES

For each view v and arm a, the effect is the average difference from the no-AI arm on the same problems:

$$\Delta^{v}_{a-N} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \big(s_{a,v}(p) - s_{N,v}(p) \big). \tag{3}$$

Its 95% interval comes from a cluster bootstrap over problems with 10,000 resamples. The four gates were fixed before the pilot:

- $\mathcal{G}_1$, practice sensitivity: $\Delta_{T-N} > 0$ on the same problem with the dialog in view.
- $\mathcal{G}_2$, crutch cost: $\Delta^{v}_{T-N} \leq -0.03$, with the upper end of the 95% interval below 0.
- $\mathcal{G}_3$, hint safety: the lower end of the 95% interval of Δ^{v}_{H-N} is above -0.03.
- $\mathcal{G}_4$, no drift (Sim_{LoRA} only): $|\Delta_{T-N}| \leq 0.03$ on the unrelated problem.

The trial reports its exam effects as absolute differences in the fraction of correct answers: -0.054 for GPT Base (95% CI $[-0.097, -0.011]$) and -0.004 for GPT Tutor ($[-0.029, +0.021]$). The -0.03 margin of $\mathcal{G}_2$ lies inside the first interval, and the lower end of the second interval almost equals the margin of $\mathcal{G}_3$. The pilot's no-AI scores (0.28–0.36) are close to the trial's control mean of 0.321, so absolute and relative versions of the target agree.

4 EXPERIMENTS

4.1 SETUP

Dataset. We use the EduardoRL transfer dataset (Macina et al., 2026). It pairs Big-Math problems (Albalak et al., 2025) with model-written near-transfer variants, and it stores the student's solve rate on each variant before tutoring (its cold solve rate). We keep the 355 problems whose variant has a cold solve rate in $[1/16, 0.75]$, which leaves out variants the student always fails or usually solves. From these we draw a seeded split (seed 2026) of 50 development and 300 held-out problems. The pilot uses the 50 development problems.

Models. The student is Llama-3.1-8B-Instruct (Grattafiori et al., 2024) in bf16. For dialogs and the frozen views, it runs on a single pinned bf16 host, because fp8 hosts change the weights. Sim_{LoRA} uses an identical copy of the weights, served with vLLM (Kwon et al., 2023) 0.30 on one A100-80GB. The tutors are Claude Haiku 5.5 at low reasoning effort. A Qwen3.6-27B judge, with reasoning off, flags dialogs in which the tutor hands over the solution.

Hyperparameters. Student turns are capped at 300 tokens. Post-test answers are sampled at temperature 0.6 with a 2,048-token cap and graded on their `\boxed{}` answer with EduardoRL's grader. We draw $K = 16$ answers per test for Sim_M and Sim_K, and 8 for Sim_C. Sim_{LoRA} uses LoRA rank 16 and $\alpha = 32$ on all seven attention and MLP projections, four SGD steps, and gradient checkpointing. Its learning rate is calibrated on the first 20 N and 20 H dialogs.

Pilot health. No student call (of 9,600) and no dialog (of 150) failed. No tutor set a new practice problem. An earlier protocol without the end rule did so in 18 of 50 T dialogs. The judge flags handover in 82% of T dialogs and 0% of H dialogs. Re-measured cold solve rates are within 0.016 of the dataset's. Dialogs and frozen post-tests cost $6.54 in API calls, plus $1.16 for Sim_C and Sim_K. Sim_{LoRA} took 22 minutes of training and 33 minutes of sampling.

4.2 RESULTS

Practice rises, as in the trial. With the dialog in view, answers raise the same-problem score by $+0.574$ $[+0.464, +0.681]$ and hints by $+0.529$. $\mathcal{G}_1$ passes: the tutors help in the moment.

No frozen simulator pays the crutch cost. Answers *raise* the score on the variant under Sim_M ($+0.122$) and Sim_C ($+0.145$; Table 1 and Figure 2). Removing copied sentences does not help. Sim_C deletes 10% of the student's sentences after answers and 15% after hints, and the gain grows. A short note (Sim_K) shrinks the gain to $+0.060$ but does not reverse it. With the full dialog in view,

Table 1: Scores on the new variant, taken without help, as differences from the no-AI arm on the same problems (95% cluster-bootstrap CI, 50 problems). $\mathcal{G}_2$ passes when $T - N \leq -0.03$ with the whole interval below 0. $\mathcal{G}_3$ passes when the lower end of the $H - N$ interval is above -0.03. The human row gives the trial's exam effects (Bastani et al., 2025).

Simulator (what it carries)	N	$H - N$	$T - N$	$\mathcal{G}_2$	$\mathcal{G}_3$
Human trial (students)	0.321	-0.004 $[-0.029, +0.021]$	$\mathbf{-0.054}$ $[-0.097, -0.011]$	target	target
Sim_M (own turns, tutor hidden)	0.331	$+0.133$ $[+0.041, +0.225]$	$+0.122$ $[+0.036, +0.209]$	fail	pass
Sim_C (Sim_M minus copied text)	0.275	$+0.168$ $[+0.065, +0.270]$	$+0.145$ $[+0.052, +0.233]$	fail	pass
Sim_K (one note, $\leq$120 tokens)	0.336	$+0.018$ $[-0.041, +0.079]$	$+0.060$ $[+0.003, +0.120]$	fail	fail
Sim_{LoRA} (weights, no transcript), $\eta = 0.1$	0.356	$+0.005$ $[-0.039, +0.050]$	$\mathbf{-0.022}$ $[-0.074, +0.028]$	fail	fail
Sim_{LoRA} (weights, no transcript), $\eta = 0.03$	0.386	-0.035 $[-0.071, +0.000]$	-0.003 $[-0.040, +0.036]$	fail	fail

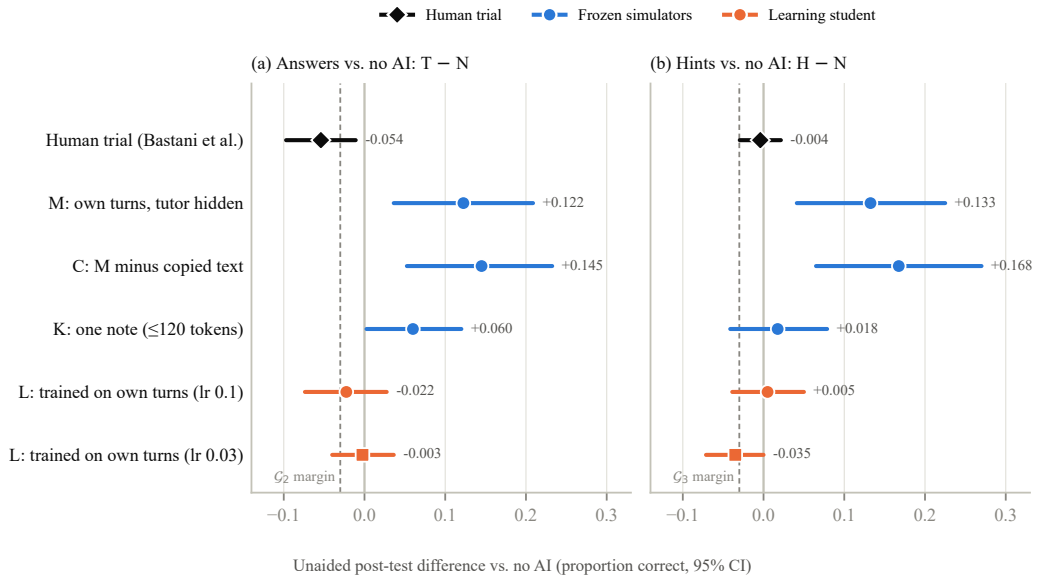


Figure 2: Effects on the new variant, taken without help, as differences from the no-AI arm (95% intervals). Frozen simulators (blue) gain from answers. The learning student (orange) removes that gain at both learning rates, but neither setting reaches the human cost (black). Dashed lines mark the -0.03 margins of $\mathcal{G}_2$ and $\mathcal{G}_3$.

the gain is $+0.315$, so most of the benefit of being told comes from the dialog in context. All three frozen simulators fail $\mathcal{G}_2$ in the wrong direction. The paired standard deviation of $T - N$ across problems is 0.32.

The learning student removes the gain but does not pay the cost. At the calibrated rate ($\eta = 0.1$), Sim_{LoRA} gives $T - N = -0.022$, the only negative estimate among the simulators. Its interval includes both zero and the human estimate, so $\mathcal{G}_2$ fails at this sample size. At $\eta = 0.03$, the estimate is -0.003. At both rates, learning in the weights removes the $+0.06$ to $+0.15$ gain of the frozen simulators, but it does not reproduce the human cost. Sim_K and Sim_{LoRA} at $\eta = 0.1$ fail $\mathcal{G}_3$ because their intervals are wide, not because hints hurt: their $H - N$ estimates are $+0.018$ and $+0.005$. At $\eta = 0.03$, the $H - N$ estimate itself turns negative (-0.035).

Where the sign comes from. Table 2 shows where the gap between answers and no AI comes from. Students without AI or with hints gain a little from their own work on the variant ($+0.016$ and $+0.021$ over the frozen model). Students given answers gain nothing (-0.006). This is the direction the generation effect predicts (Slamecka & Graf, 1978).

Three findings weaken this result. First, the amount of practice does not explain the gain. Students given answers wrote more of their own tokens than students without AI (median 570 vs. 535), and across all dialogs the rank correlation between trained tokens and gain is -0.03. Whatever Sim_{LoRA}

Table 2: What Sim_{LoRA} learned: the adapted student minus the frozen model on the same tests (95% CI), at learning rate 0.1. Trained tokens and loss drops are medians over dialogs.

Arm	Trained tokens	Loss drop	Variant	Same problem	Unrelated problem
N (no AI)	535	40.2%	$+0.016$ $[-0.024, +0.056]$	-0.028 $[-0.087, +0.030]$	-0.043 $[-0.110, +0.022]$
H (hints)	761	41.7%	$+0.021$ $[-0.025, +0.066]$	-0.058 $[-0.113, -0.005]$	-0.030 $[-0.095, +0.033]$
T (answers)	570	37.6%	-0.006 $[-0.048, +0.036]$	-0.060 $[-0.125, +0.005]$	$+0.022$ $[-0.030, +0.072]$

learns depends on what the students wrote, not on how much. Second, training lowers same-problem scores in every arm (-0.03 to -0.06), and adapted answers hit the token cap more often (13–15% vs. 9–12% for the frozen views). Both are signs of overfitting to the dialog. Third, $\mathcal{G}_4$ fails: on the unrelated problem, $T - N = +0.065$ $[-0.005, +0.133]$. Training shifts students in general, not only on what they practiced.

Learning-rate sensitivity. The calibration rule had to choose between two rates that sit almost equally far from its target. The median loss drop is 7.0% at $\eta = 0.03$ and 42.5% at $\eta = 0.1$, which are 0.180 and 0.175 away from the 25% target. Rates of 0.3 and above diverged. An earlier attempt, before we added gradient checkpointing, picked 0.03 by the same margin. We therefore reran Sim_{LoRA} with η fixed at 0.03 (median loss drop 6–7%). At this rate, students without AI and students given answers both gain about $+0.04$ over the frozen model on the variant ($+0.041$ and $+0.039$). As a result, $T - N$ vanishes (-0.003 $[-0.040, +0.036]$) and $\mathcal{G}_4$ passes ($+0.025$ $[-0.037, +0.087]$), but students given hints fall behind ($H - N = -0.035$ $[-0.071, +0.000]$). The human-sign estimate at $\eta = 0.1$ therefore depends on the learning rate, while the removal of the frozen simulators' gain does not.

Limitations. The pilot has 50 problems and one dialog per arm, so its intervals are wide. It uses one student model, a different tutor model from the trial's GPT-4, and Big-Math problems rather than the trial's curriculum. The human target is an intention-to-treat effect over all students with access, not all of whom copied, so the cost for students who copy is likely larger. Finally, Sim_C overlaps with the handover judge by construction, and the judge's 82% handover rate is close to the protocol's 80% gate.

5 Conclusion

We built a test that scores student simulators against the treatment effects of a human trial. Simulators that carry the dialog as text fail it in the wrong direction: being told the answer helps them on a new variant, where it hurt students. A student that learns only from its own tokens removes this gain at both learning rates we tried, but it does not reproduce the human cost. Its estimate has the human sign at one rate and is zero at the other. Three next steps follow. The first is the 300-problem held-out run, with two dialogs per arm and the try-first (S) and confidently wrong (W) arms; its harness is built and has 77–86% power to detect the human effect at the pilot's variance. The second is replication with other student models. The third is training tutors with RL against Sim_M and Sim_{LoRA}, to test whether the choice of simulator changes the policy that is learned.

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